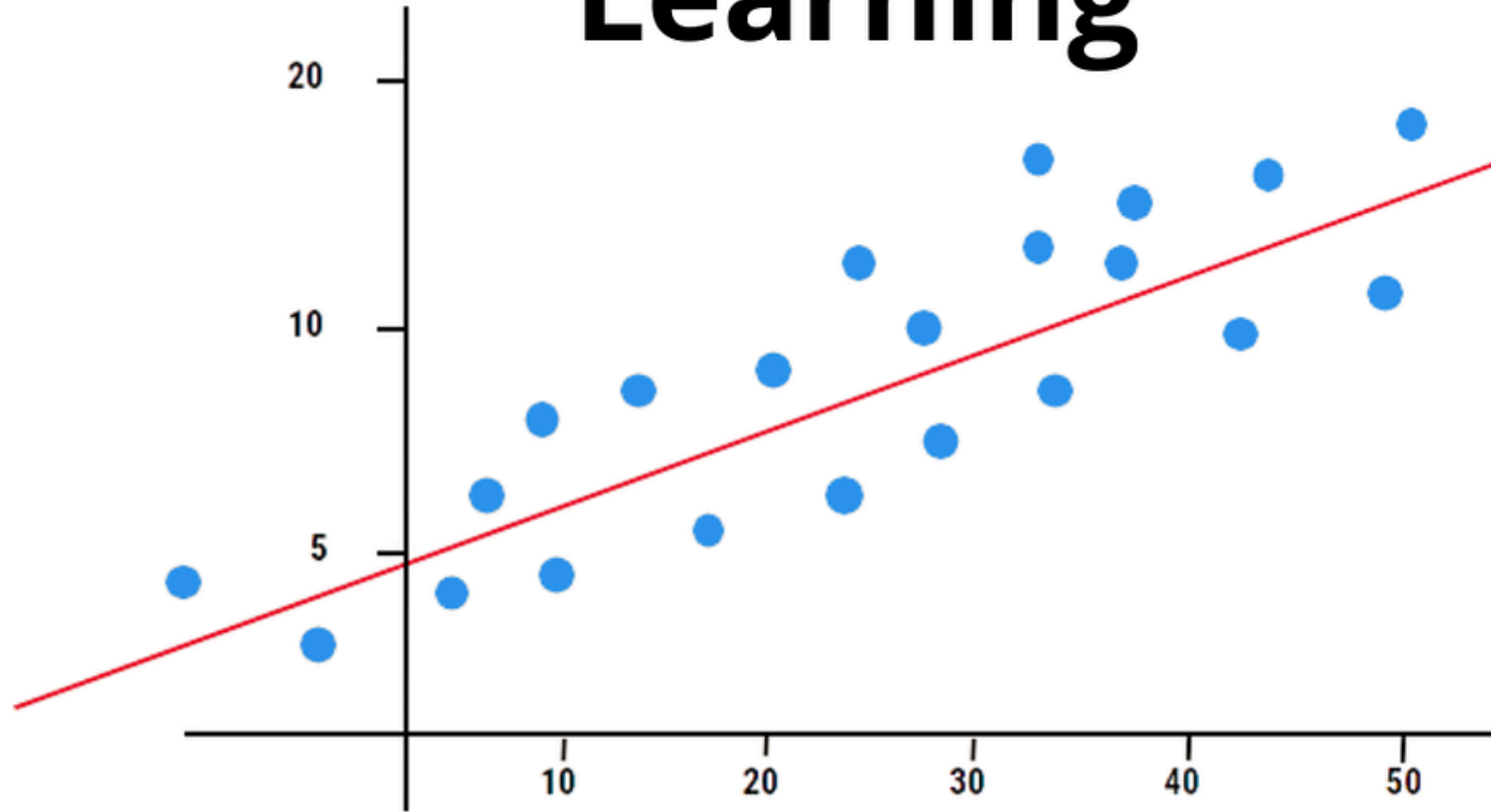
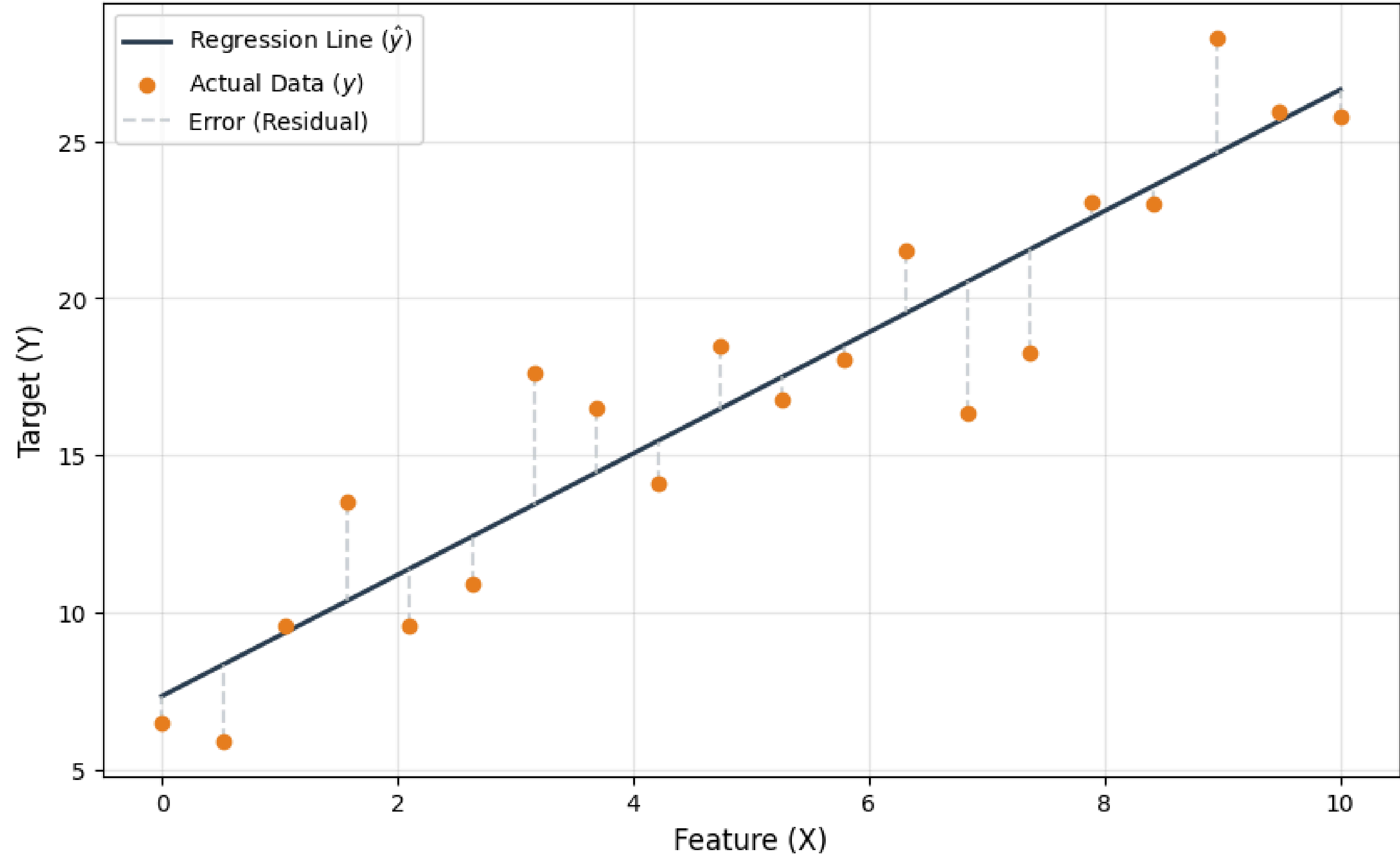


Loss Function in Machine Learning



Visualizing Errors in Linear Regression



What is a Loss Function?

A loss function measures how wrong your model's prediction is.

👉 If prediction is good → small loss

👉 If prediction is bad → big loss

The model's goal during training is:
minimize the loss

Because the model must know:

“How bad am I right now?”

Without loss, the model has no direction to improve.

The optimizer (like Gradient Descent or Adam) uses the loss value to update weights.

Flow:

How Loss Works in Practice

Suppose real value = 10

Model predicted = 7

Loss function computes difference → tells model it's wrong by 3.

Now weights will change to reduce that error next time.

Regression Metrics (with formula)

Let

y_i = actual value

$\hat{y}_i$ = predicted value

n = number of data points

1) Mean Absolute Error (MAE)

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Explanation:

It measures the absolute difference between actual and predicted values.

Why used:

Because it is easy to understand and gives error in the same unit as the output. It is also less affected by very large errors.

2) Mean Squared Error (MSE)

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Explanation:

It squares the difference between actual and predicted values.

Why used:

Large errors get more punishment, so the model tries harder to avoid big mistakes. It is commonly used during training.

3) Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Explanation:

It is simply the square root of MSE.

Why used:

After taking the root, the value comes back to the original unit, which makes interpretation easier while still penalizing large errors.

4) R^2 Score (Coefficient of Determination)

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

where $\bar{y}$ is the mean of actual values.

Explanation:

It tells how well the model fits the data.

Why used:

It is useful for comparing models. A value closer to 1 means better performance.

Always default to MSE for regression training. It's smooth, easy to optimize, and penalizes big errors.

Model Validation and Evaluation-Confusion Matrix

❖ The confusion matrix is a matrix used to determine the performance of the classification models for a given set of test data.

		Predicted	
		Negative (N) -	Positive (P) +
Actual	Negative -	True Negative (TN)	False Positive (FP) Type I Error
	Positive +	False Negative (FN) Type II Error	True Positive (TP)

❖ **True Negative (TN):** Model has given prediction No, and the real or actual value was also No.

❖ **True Positive (TP):** The model has predicted yes, and the actual value was also true.

❖ **False Positive (FP):** The model has **predicted Yes, but the actual value was No**. It is also called a **Type-I error**.

❖ **False Negative (FN):** The model has **predicted No, but the actual value was Yes**, it is also called as **Type-II error**.

Prepared By: Er. Sudarshan Subedi

Model Validation and Evaluation-Confusion Matrix

Classification Metrics

1. **Classification Accuracy:** It is one of the important parameters to determine the accuracy of the classification problems. It defines how often the model predicts the correct output.

$$\text{Accuracy} = \frac{TP+TN}{TP+FP+TN+FN}$$

2. **Misclassification Error:** It is also termed as Error rate, and it defines how often the model gives the wrong predictions. The value of error rate can be calculated as the number of incorrect predictions to all number of the predictions made by the classifier.

$$\text{Error} = 1 - \text{Accuracy} = \frac{FP+FN}{TP+FP+TN+FN}$$

Model Validation and Evaluation-Confusion Matrix

Classification Metrics

3. **Precision(exactness):** what % of tuples that the **classifier labelled as positive are actually positive.**

$$\text{Precision} = \frac{TP}{TP + FP}$$

4. **Recall/Sensitivity/True Positive rate(TPR):** Recall or Sensitivity is the **Ratio of true positives to total (actual) positives in the data.** what % of positive tuples did the classifier labelled as positive.

$$\text{Recall} = \frac{TP}{TP + FN}$$

Model Validation and Evaluation-Confusion Matrix

Classification Metrics

7. False Positive Rate(FPR): The false positive rate is calculated as the ratio between the number of negative events wrongly categorized as positive (false positives) and the total number of actual negative events (regardless of classification).

$$FPR = \frac{FP}{N} = \frac{FP}{TN+FP}$$

7. False Negative Rate(FNR)

$$FNR = \frac{FN}{P} = \frac{FN}{TP+FN}$$

Model Validation and Evaluation-Confusion Matrix

Classification Metrics-Example

Consider the following confusion matrix.

n=165	Predicted: NO	Predicted: YES
Actual: NO	50	10
Actual: YES	5	100



n=165	Predicted: NO	Predicted: YES	
Actual: NO	TN = 50	FP = 10	60
Actual: YES	FN = 5	TP = 100	105
	55	110	

$$\text{Accuracy} = \frac{TP+TN}{TP+FP+TN+FN} = \frac{100+50}{100+10+50+5} = \frac{150}{165} = 0.9090$$

$$\text{Error} = 1 - \text{Accuracy} = 0.090$$

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{100}{100+10} = 0.9090$$

Model Validation and Evaluation-Confusion Matrix

Classification Metrics-Example

Consider the following confusion matrix.

n=165	Predicted: NO	Predicted: YES
Actual: NO	50	10
Actual: YES	5	100



n=165	Predicted: NO	Predicted: YES	
Actual: NO	TN = 50	FP = 10	60
Actual: YES	FN = 5	TP = 100	105
	55	110	

$$\text{Recall/Sensitivity/True Positive rate(TPR)} = \frac{TP}{TP+FN} = \frac{100}{100+5} = 0.9523$$

$$\text{Specificity/TNR} = \frac{TN}{TN+FP} = \frac{50}{50+10} = 0.833$$

$$\text{FPR} = \frac{FP}{TN+FP} = 10/(50+10) = 0.1667$$

Confusion Matrix Numerical

An input sequence “A A B B B A A A B B” was used for classification. The classifier ‘X’ predicted the sequence as: “A A A A B B A A A B”. Developing the corresponding confusion matrix for the classifier and find their corresponding:

1. Accuracy
2. Precision
3. True Positive Rate
4. False Positive Rate

your task

